

LESSON 312a (1.0 GROUND)

Lesson Objective

This lesson is a ground review conducted by the instructor in preparation for the stage check to ensure student proficiency and confidence in instrument flight.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

Lesson Content

Pilot Qualifications

- Instrument Rating Privileges (61.3e):
 - Ask the student what they can do with an instrument rating.
- IFR Flight Plan and Clearance Required (91.173):
 - When is an IFR flight plan required?
 - Is filing IFR enough to fly through IMC?
 - Is it legal to fly in uncontrolled airspace in IMC without filing?
- Currency as an Instrument Pilot (61.57):
 - How long do instrument privileges last after the practical test?
 - What must be done to retain currency?
 - What must be logged to maintain currency?
- Safety Pilots (91.109):
 - When must a safety pilot be used?
 - Who can be a safety pilot?
 - Can a Private Pilot be a safety pilot in an Arrow without a complex endorsement?
 - Can a Private Pilot be a safety pilot in a Seminole without AMEL?
 - Does the safety pilot need to be current?
 - What type of flight controls must the airplane possess to use a safety pilot?
 - What if the safety pilot cannot see well enough?
 - What kind of time may be logged by the safety pilot? The instrument pilot?
- Conditions Required to Log Tasks for Currency (61.57):
 - Ask the pilot in training what conditions are required to log the approaches, etc.
 - Define actual instrument conditions.
 - Can a safety pilot be brought into IMC if the pilot flying is not current?
- Instrument Proficiency Check (61.57):
 - When are IPCs required?
 - Who may administer an IPC?
 - What tasks are involved in an IPC?
 - Can an IPC be administered in a simulator?

Weather Information

- Sources of Weather Data for Flight Planning Purposes.
 - Sources that fulfill the legal preflight requirements:
 - Calling a weather briefer.
 - 1800wxbrief.com.
 - ForeFlight generated weather briefing.
 - Aviation Weather Center.
- Acceptable Weather Products and Resources.
 - METAR:
 - [AIM 7-1-28 and 7-1-29](#) details the shorthand used in various ICAO standardized weather reports.
 - [Live METARs](#).
 - [Info page on AWC](#).
 - What is a METAR?
 - How long is a METAR valid for?
 - How often is a METAR issued?
 - What is a SPECI?
 - TAF:
 - [Live TAFs](#).
 - [Info page on AWC](#).
 - What is a TAF?
 - How long is a TAF valid for?
 - How often is a TAF issued?
 - Winds/Temps Data:
 - [AIM 8-3-1](#) details FB Winds.
 - [Live FB Winds](#).
 - [Info page on AWC](#).
 - What are FB Winds?
 - How often are FB Winds issued?
 - How far out does each issuance forecast?
 - How long is each issuance valid for?
 - PIREP:
 - [AIM 7-1-18,20,21,22,23](#).
 - [Live PIREPs](#).
 - What is a PIREP?
 - How often are they issued?
 - Prognostic charts:
 - Low-Level Significant Weather Chart (Surface to FL240):
 - [Info page on AWC](#).
 - Quiz the student on the information shown on a prognostic chart.

- Surface Prognostics Chart:
 - [Info page on AWC.](#)
 - Quiz the student on the information shown on a surface analysis chart.
- Freezing level chart:
 - [Info page on AWC.](#)
- Graphical Forecast for Aviation (GFA) tool:
 - [Info page on AWC.](#)
- AIRMETs:
 - [Info page on AWC.](#)
 - What is an AIRMET?
 - What is the difference between an AIRMET and a G-AIRMET?
 - How often are they issued, and how long are they valid for?
 - What are the different types of G-AIRMETs?
 - What condition must be met for a G-AIRMET to be issued?
- SIGMETs:
 - [Info page on AWC.](#)
 - What is a SIGMET?
 - How often are they issued, and how long are they valid for?
 - What condition must be met for a SIGMET to be issued?
- Convective SIGMETs:
 - [Info page on AWC.](#)
 - What is a convective SIGMET?
 - How often are they issued, and how long are they valid for?
 - What conditions must be met for a convective SIGMET to be issued?
- Convective Outlook:
 - What is a convective outlook?
 - How often are they issued, and how long are they valid for?
- Center Weather Advisory:
 - What is a center weather advisory?
 - Who issues them?
 - When are they issued?
 - How long are they valid for?
- Meteorology:
 - Atmospheric composition and stability:
 - What is atmospheric stability?
 - What is the general composition of the atmosphere?
 - Does the composition of the atmosphere change with altitude?
 - What is an air mass?
 - What is a front?
 - What is a cold front? What are the characteristics of a cold front?

- What is a warm front? What are the characteristics of a warm front?
- What is a stationary front? What are the characteristics of a stationary front?
- What is an occluded front? What are the characteristics of an occluded front?
- What is frontogenesis? What about frontolysis?
- Wind:
 - How does air flow relative to the isobars at the surface? And aloft?
 - What does a tight pressure gradient indicate in terms of wind?
- Temperature:
 - What is the temperature lapse rate on a standard day?
 - What is a temperature inversion?
 - When might a temperature inversion be experienced?
- Moisture/precipitation:
 - What are the different intensities of precipitation?
- Weather system formation, including air masses and fronts:
 - What is the source of all weather?
- Clouds:
 - What are some types of clouds?
 - What counts as a ceiling?
- Turbulence:
 - Refer to [TBL 7-1-10](#) in [AIM 7-1-21](#).
 - What is the difference between turbulence and chop?
 - Turbulence
 - Turbulence is a random, sporadic movement of the aircraft.
 - Chop
 - Chop is a rapid and somewhat rhythmic movement of the aircraft.
 - What are the intensities of turbulence or chop?
 - Light
 - Moderate
 - Severe
 - Extreme
 - What are the frequencies of turbulence?
 - Occasional
 - Intermittent
 - Continuous
- Thunderstorms and Microbursts:
 - What are the criteria required for the formation of a thunderstorm?
 - What are the stages of a thunderstorm?
 - Different types of thunderstorms:
 - Air-mass.
 - Steady State.

- Squall line.
- Embedded.
- Single cell.
- Multi-cell.
- Supercell.
- What weather to expect in a thunderstorm:
 - Lightning.
 - Turbulence.
 - Hail.
 - Tornadoes.
 - Wind shear.
 - Icing.
- Icing and freezing level information:
 - [AIM 7-1-19 and 7-1-20.](#)
 - What is the difference between known icing and known icing conditions?
 - What are the performance impacts of icing?
 - What are the different categories of icing in general?
 - Structural.
 - Induction.
 - What can the pilot do when dealing with induction icing?
 - Instrument.
 - Look for more than just pitot/static icing.
 - What are the different kinds of structural icing?
 - What conditions are necessary for the formation of structural icing?
 - What are the different levels of icing rates?
 - What is the difference between an anti-ice and de-ice device?
- Fog/mist:
 - What is the difference between fog and mist?
 - How does fog form?
 - What are the different types of fog?
- Frost:
 - What is frost?
 - How does it form?

Aircraft Systems and Aircraft Flight Instruments Related to IFR Operations

- Required Equipment - 91.205 (d)(e), 91.131(c)(1), 91.215, 91.411 and 91.413.
 - What equipment is required to fly IFR (day and night)?
 - What inspections and checks are required to fly IFR?
 - Ask the student various questions regarding the pieces of equipment that they listed (e.g, manifold pressure gauge, temperature gauge, alternator vs. generator, source of power,

etc.)

- Pitot-Static System:
 - Ask the student to explain/draw the pitot-static system.
 - Airspeed indicator:
 - Discuss how the airspeed indicator works.
 - What if, during a climb, airspeed increases?
 - What if, during a descent, airspeed increases?
 - Ask what to do if there is a blockage.
 - Review the types of airspeed.
 - Altimeter:
 - Discuss how the altimeter works.
 - What is the failed indication?
 - Review the types of altitudes.
 - Vertical speed indicator:
 - Discuss how the vertical speed indicator works.
 - What is the failed indication?
 - Is the aircraft IFR legal if the VSI reads +200 fpm on the ramp?
 - Review the effects of alternate static on the instruments and the procedure to use it.
- Vacuum system:
 - Ask the student to explain/draw the system
 - Where does the air come from?
 - What does the vacuum air do to the instruments?
 - What if the system fails? What is the indication?
 - Do we have a vacuum system on our airplane?
- Gyroscopic instruments:
 - Ask what the gyroscopic principles are.
 - Attitude indicator:
 - What is the operating principle?
 - How does the gyro line up with the horizon?
 - What is the power source for the instrument in the airplane?
 - If the training airplane's attitude indicator is electrical, when does the battery come on?
 - How long does this battery last?
 - What are errors associated with the attitude indicator?
 - Ask the student to describe the instrument's limitations.
 - Heading indicator:
 - What is the operating principle (for the training aircraft)
 - For the glass systems, what keeps the instrument aligned to magnetic north?
 - Why does the mechanical version drift?
 - What are the mechanical instrument's limitations?

- Turn coordinator:
 - What is the instrument's operating principle?
 - What is its power source?
 - How is this instrument different from a turn and slip indicator?
 - What is its failed indication?
- Electrical system:
 - Ask the student to describe the batteries, alternators, including backups in terms of amps, amp-hours, and volts.
 - Have the student explain how electricity flows from the battery to the main bus.
 - How is the electrical system protected?
 - Electrical failure:
 - Ask the student to describe when the standby alternator functions.
 - Review the checklist procedure for the standby alternator.
 - Discuss load shedding as per the checklist.
 - Ask how one knows when the alternator is overloaded and why that condition arises.
- Autopilot (S-TEC 55X):
 - How does the autopilot work?
 - What are the servos, and where are they?
 - How can the autopilot be disconnected?
 - What is the difference between the flight director and autopilot?
 - What does Avidyne recommend if the PFD fails in IMC?
- Avidyne system:
 - PFD:
 - Have the pilot explain its limitations.
 - What components are inside the PFD?
 - Ask the student to explain the ADAHRS.
 - How does the ADC work? What information does it provide?
 - How does the AHRS work? What information does it provide?
 - Ask about abnormal operations.
 - MFD:
 - Have the pilot explain its limitations.
 - What is the DAU?
 - When may the MAP page be displayed?

Navigation Equipment

- VOR:
 - Ask the student how a VOR works.
 - What is the frequency range for VORs?
 - What are the different types of VORs?
 - What are the different service volumes?

- What is the MON?
- How often is the inspection required?
- What are the different ways of testing the VOR?
- What must be recorded?
- What are the limitations of the VOR?
- DME:
 - How does the DME Work?
 - How accurate is the DME?
 - Do we have a DME?
 - What are the limitations of the DME?
- ILS:
 - Localizer:
 - Have the student explain/draw the function and service volume of the localizer.
 - What are the frequencies?
 - What is the purpose of a localizer?
 - Within how many degrees must a localizer be aligned with the runway centerline?
 - What is a localizer back course?
 - Glideslope:
 - Have the student explain/draw the function and service volume of the glideslope.
 - What are the frequencies?
 - What is the purpose of a glideslope?
 - What is a false glideslope?
 - Marker beacons:
 - What are marker beacons used for?
 - Have the student identify each beacon, its color, and morse code.
 - Approach lights:
 - What are approach lights used for?
 - What are the different types?
- GPS:
 - Principle of operation:
 - Ask how GPS functions to derive location information.
 - How many satellites are required to define a fix?
 - How many satellites are required for IFR?
 - Components:
 - Ask the student to describe the 3 segments of the GPS system.
 - Modes of operation:
 - How are GPS modes annunciated?
 - Ask the student to describe the sensitivities of the different GPS modes.
 - Capabilities
 - How do we know if a GPS unit can be used for IFR operations?

- Does the GPS database need to be current to file IFR?
- Does the database need to be current to substitute NDB/VOR/DME?
- RAIM:
 - What is the purpose of RAIM?
 - How does RAIM work?
 - What is the difference between fault detection and fault exclusion?
 - Do you need to check for RAIM prior to flying IFR?
 - Is RAIM necessary with WAAS equipment?
 - What should be done if RAIM is lost en route?
- WAAS:
 - What is the purpose of WAAS?
 - How does WAAS work?
 - How do you know if WAAS is on/available?
 - Is backup equipment needed with WAAS?
- GBAS:
 - What is the purpose of GBAS?
 - How does GBAS work?
 - How does it differ from WAAS?
 - What does GBAS enable a pilot to do (GLS approach)?

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 312b.
- Assign Homework:
 - Review stage 1 instrument procedures!

Completion Standards

This lesson is complete when the student demonstrates proficiency in all line items.